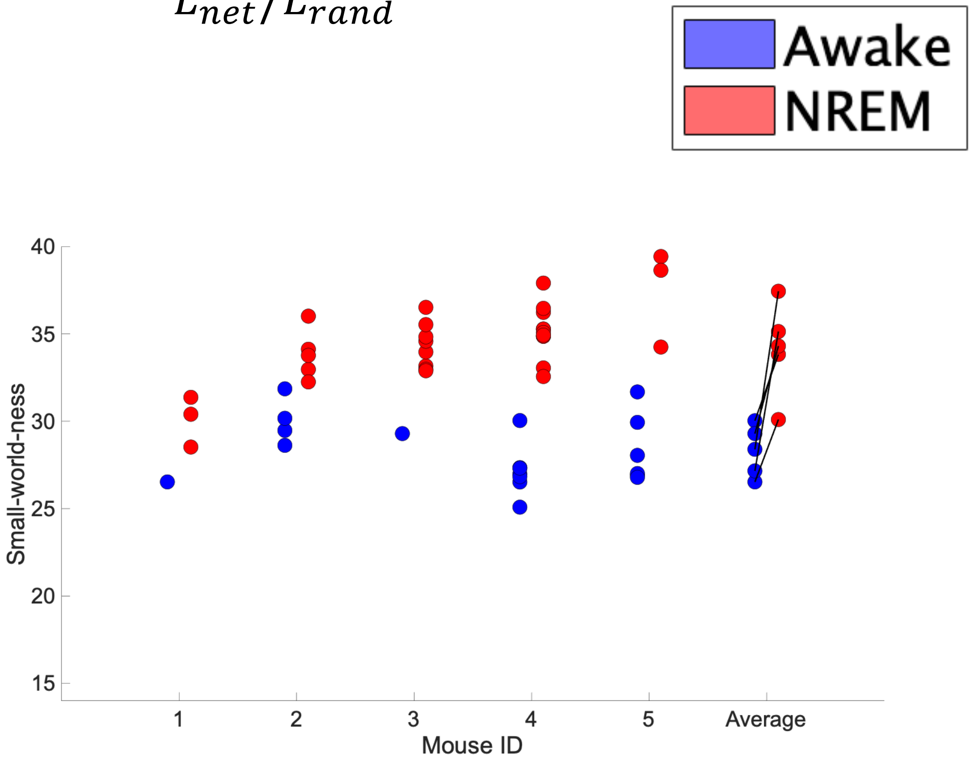
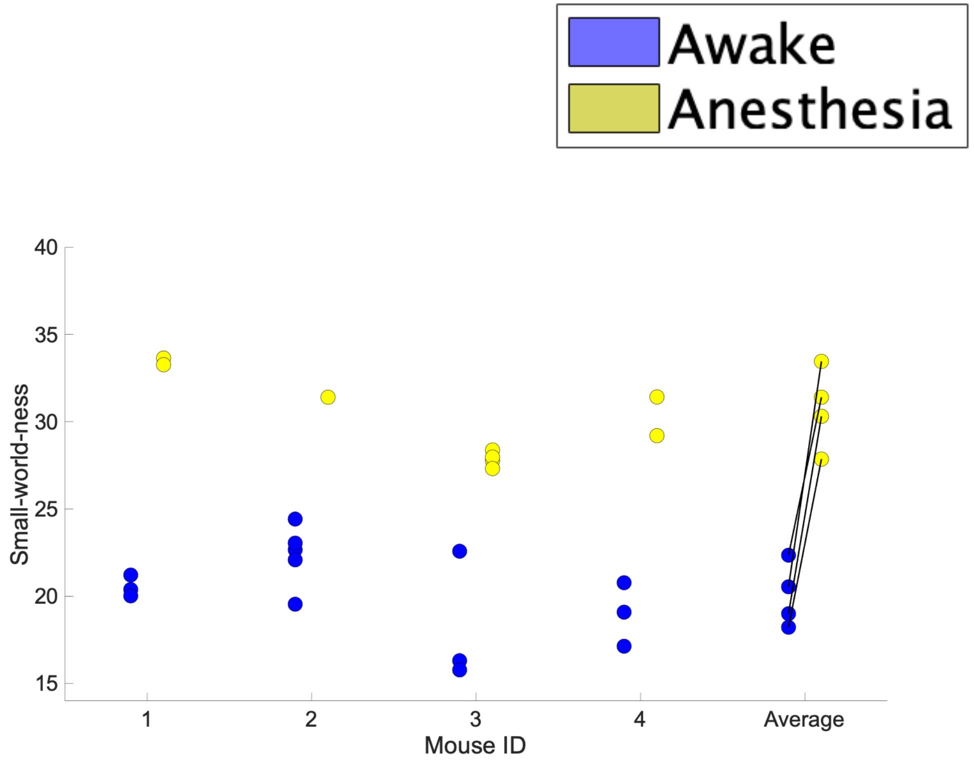
Small-worldness: Network integration (information flow)

- 睡眠・麻酔時の方が、覚醒時よりもsmall-worldnessが高い

$$SMN = \frac{C_{net}/C_{rand}}{L_{net}/L_{rand}}$$

Balance between path length and clustering coefficient

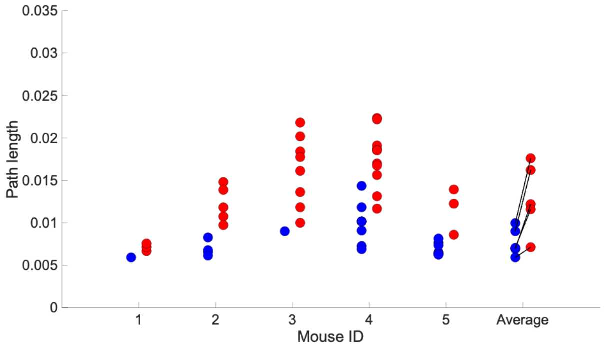




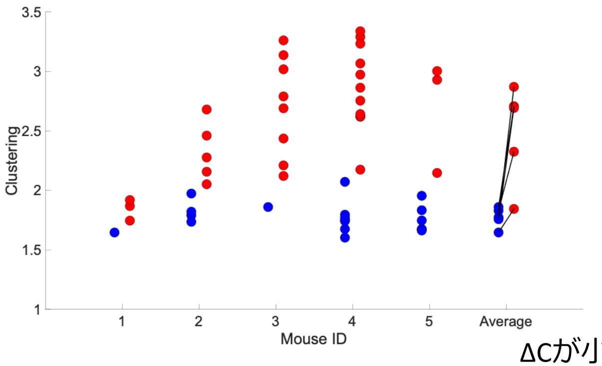
Long path length and high clustering coefficient in the unconscious state

$$\Delta C = \frac{C_{reg} - C_{net}}{C_{reg} - C_{rand}},\ \Delta L = \frac{L_{net} - L_{rand}}{L_{reg} - L_{rand}}$$

ΔL

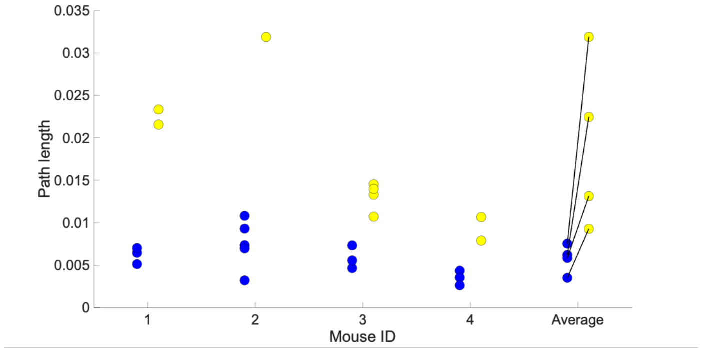


$\frac{1}{\Delta C}$

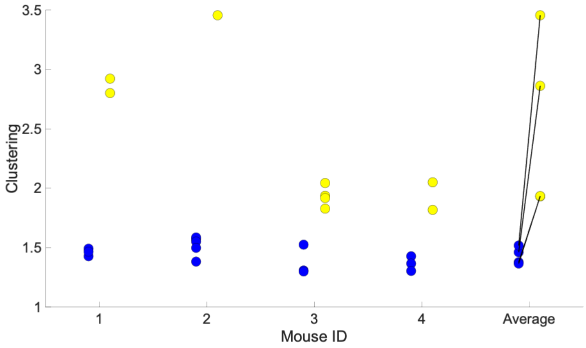


ΔCが小さいほどクラスタリング係数が大きい
1/ ΔCが大きいほど局所化している

ΔL



$\frac{1}{\Delta C}$

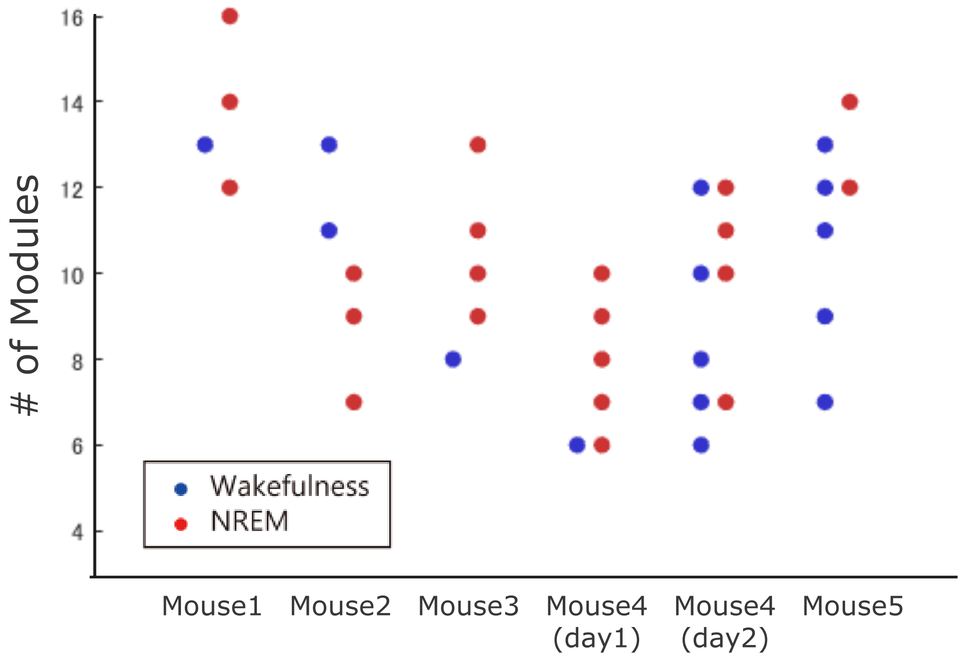


Conscious state: shorter path length and lower clustering coefficient
→The awake brain exhibits stronger global integration, enabling more efficient information transfer across the network.

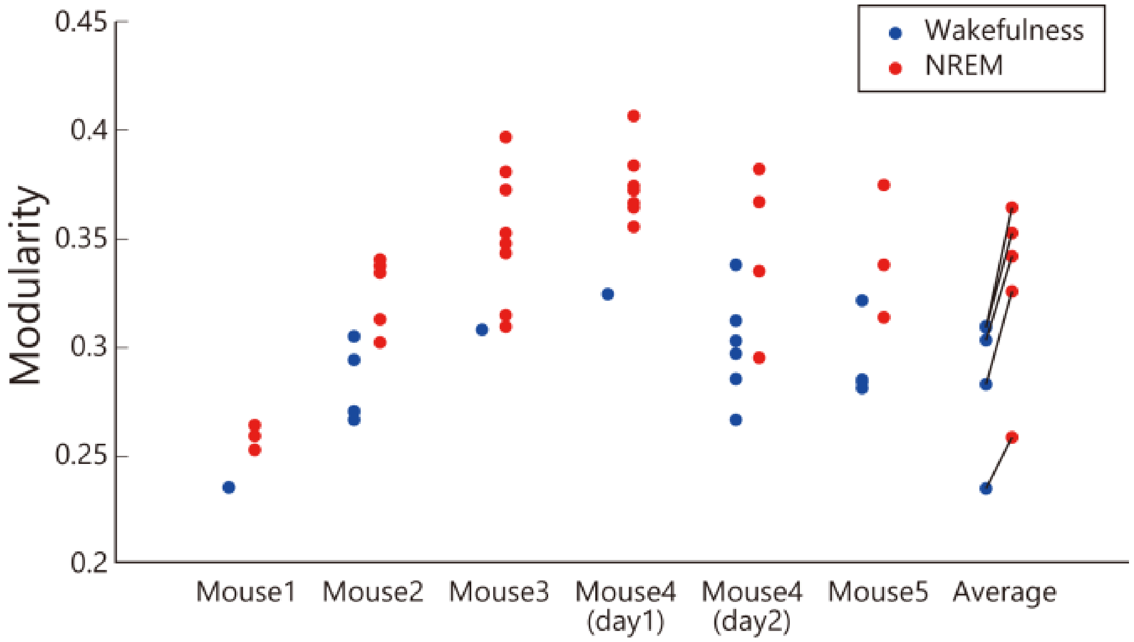
Unconscious state: longer path length and higher clustering coefficient
→Communication becomes more localized.

Does increased local clustering lead to stronger modular organization?

Modularity (Functional segregation)

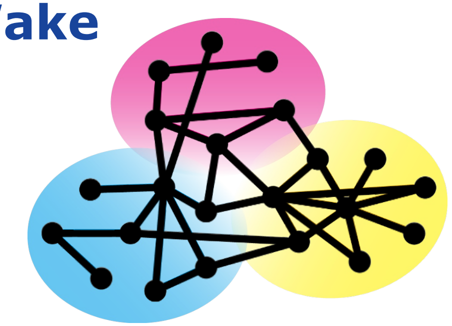


- モジュールの数はWake・NREM間で差はない。

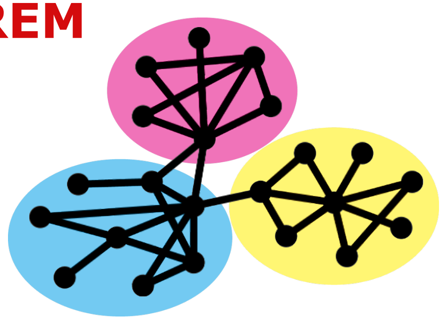


- NREMの方がモジュール性が高い。

Wake



NREM

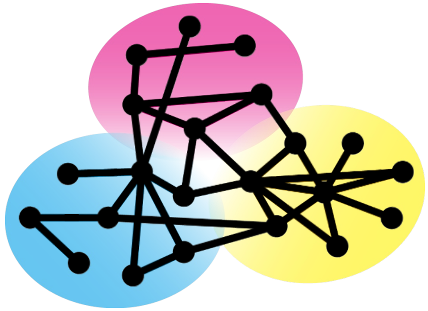


⇒モジュールの数は状態間で差がないが
NREMネットワークの方が**より分断している**

分断：fMRIなどによるマクロな観察結果と一致。

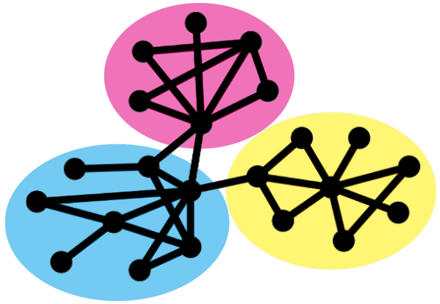
State-Dependent Reorganization of Functional Networks

Conscious state
(Wakefulness)



Integrated

Unconscious state
(NREM and Anesthesia)



Segregated

- Localという単語はここでは入れない！！
- Within/betweenで表現

Conscious state		Metric	Unconscious state	
↓	(Efficient global communication)	**Path length (*L*)**	↑	(Reduced global communication)
↓	(Less between connectivity)	**Clustering Coefficient (*C*)**	↑	(Stronger within connectivity)
↓	(Weaker segregation between modules)	**Modularity (*Q*)**	↑	(Stronger segregation between modules)

More globally integrated network with efficient information transfer.

More localized and modular network with reduced information transfer across modules.